

Design and control of a dual-stage feed drive

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Abstract

High precision positioning over a large workspace is a fundamental feature of a precision machine. Connecting coarse (large stroke) and fine (high resolution) drive stages, in series, to form a dual-stage feed drive (DSFD) system can provide the desired performance. The DSFD concept has applications that include fast tool servos for the creation of asymmetric surfaces or online chatter suppression, and micro–macro robots for high precision assembly. This paper studies the design of DSFDs for machine tools. The design issues are discussed with special considerations for the dynamics and control of the two drive stages. Two DSFDs, single-axis and two-axis, are designed with piezoelectric actuators (PAs) for the fine stages and linear motors (LMs) for the coarse stages. Both feature flexures for frictionless precision motion that are designed to meet the static and dynamic requirements of a milling process. A model-based control algorithm ensures that the stages work together in a complementary fashion. The single-axis DSFD reduced the tracking error by about 75% in comparison to a similarly controlled LM drive. A second DSFD was built for milling experiments. In sinusoidal profile cutting the maximum tracking error was reduced by 83% and the average magnitude of the error was reduced by 63%. In sharp corner cutting the DSFD reduced the maximum tracking error by 38% and the average magnitude of the error by 39%.

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1. Introduction

High precision positioning is one of the most important features of a precision machine [1]. Such a machine is required to provide versatility, speed and workspace as well as high precision positioning. Combining coarse (large stroke) and fine (high resolution) drive stages, connected in series, to create a dual-stage feed drive (DSFD) provides the capacity of a large workspace with the property of high precision motion [2].

The DSFD concept has several applications in manufacturing, robotics and data storage. The majority of DSFD applications are in hard disks. The industry continues to strive for increased storage densities and reduced data access times. This necessitates performance improvements of the head positioning system in terms of fast transition

from one track to another (track seeking), fast and accurate settling, and precise track following of the target track. To meet these requirements, the servo bandwidth of the head positioning system must be increased to lower the sensitivity to disturbances such as disk flutter vibrations, spindle motor run-out, windage, and external vibration. The servo bandwidth, however, is mainly limited by the mechanical resonance of the head positioning mechanism. Several DSFDs have been developed that can increase the servo bandwidth [3–7].

Micro–macro robots are another example of DSFDs that provide advantages when both large work space and precise end-effector positioning are required [8]. The parallel coupled micro–macro robot system, described in Ref. [9], exhibits superior characteristics when compared to typical single actuator robot systems. The benefits of this system are improved small signal force control bandwidth, reduced impedance, reduced distortion, reduced impact forces, improved small signal position bandwidth, improved force resolution and dynamic range.

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Nomenclature

B_M	LM damping coefficient	M_P	PA moving mass
B_P	PA damping coefficient	R_{LM}	LM receptance
C_M	LM feedback controller	T	sampling period
C_P	PA feedback controller	T_{LM}	LM transmissibility
e	positioning error	u_M	control command to LM
e_{PA}	PA error	u_P	control command to PA
$e_{tracking}$	tracking error	v	velocity
e_{ss}	steady-state error	z	forward shift operator
$e_{threshold}$	error threshold	r	position setpoint
F_c	cutting force	r_{LM}	LM setpoint
F_M	LM force	r_{PA}	PA setpoint
F_P	PA force	y	total position
F_T	transmitted force between PA and LM	y_{LM}	LM position
F_{TB}	transmitted force to machine base	y_{PA}	PA position
i_M	LM current	$\ddot{y}_{LM}$	LM acceleration
K_A	LM amplifier gain	$\dot{y}_{LM}$	LM velocity
K_F	LM force constant	$\ddot{y}_{PA}$	PA acceleration
K_P	PA stiffness	$\dot{y}_{PA}$	PA velocity
M_M	LM moving mass		

A fast tool servo for machining is a direct use of the concept and its applications range from the creation of asymmetric surfaces to online chatter suppression [10]. Fast tool servos have been applied to the diamond turning of brittle materials [11], where the metal removal rate has to be kept in the material ductile zone to achieve high surface finish. Brittle materials such as glass, ceramics and germanium have considerable research applications in optics and precision engineering systems. Fast tool servos provide an independently operated positioning device that has a high bandwidth and accuracy compared to a conventional ball screw drive. Because machine tools are traditionally massive, the inertia involved makes high-speed actuation of the slide ways impractical [12]. Using a fast tool servo as a secondary actuation device in the direction of interest should improve the overall performance of the machine tool. Fast tool servos also can be used to correct machine related errors such as spindle errors and parasitic vibrations [13]. They have also been widely used for the creation of asymmetric surfaces on a component, such as mirrors, anamorphic optics, and generalized surfaces [10,13]. The only published applications of fast tool servos have been with turning. Fast tool servos are also typically controlled independently of the original drive.

This paper presents the design and implementation of DSFDs for machine tools. Unlike fast tool servos, the proposed designs are for the entire feed drive rather than an addition to it. This has the advantage of allowing us to design the controller such that the two drives work together in a complementary fashion. The paper is organized as follows. The first section presented the motivation for the work and a general background. Section 2 explains the limitations of single-stage feed drives for precision

positioning. Section 3 introduces the issues involved in the design of a DSFD. Section 4 presents the design and implementation of a single-axis DSFD system. In Section 5 the design for a two-axis DSFD is described. Section 6 presents the experimental results with a DSFD while performing metal cutting. Section 7 concludes the paper.

2. Limitations of single-stage feed drives

Ball screw drives are dominant as feed drives but several limitations are associated with them. Pritschow and Philipp [14,15] and Pritschow [16] summarized conventional ball screw feed drive limits of performance to be: transmission errors, dead zone, backlash, elasticities, large inertia, and wear. Ball screw feed drives are also limited dynamically by their low first natural frequency. This represents a major drawback for high precision systems, where the feed drive has to show high dynamic stiffness against external excitations.

LMs represent a good choice for high precision feed drives. They are systems with higher natural frequencies and higher stiffness because there are no elasticities involved. However, LMs possess some limitations when it comes to submicron positioning. Force ripple in LMs [17], stick-slip friction and quantization error could affect the precision, especially with high resolution positioning.

Positioning systems based on PAs have received increased attention recently in many high precision applications. They have advantages that include: unlimited resolution, large force, fast expansion and no magnetic effects [18]. Their nano-resolution makes PAs suitable for

applications that require high precision and nano-resolution positioning such as:

- Precision machining in metals, optics, laser cutting and diamond turning.
- Metrology, laser systems and optical inspection.
- Micro-manufacturing, assembly and testing.
- Telecommunication, photonics, fiber optics and fiber alignment.
- Microscopy.
- Data storage.
- Semiconductor, LCD and flat panel manufacturing.
- Medical technology.
- Aerospace, defense technology and imaging.

Efforts have been made to increase the travel range of PAs. Presently, the positioning range is limited to couple of hundred microns. This relatively large range requires the use of large number of piezoelectric layers. This reduces the actuator bandwidth dramatically because the many layers act as a large capacitor that leads to saturation of the high voltage amplifier [19]. The limitations of a single-stage feed drive can be overcome by combining multiple, usually dual, feed drive stages and adopting the merits of each drive stage.

3. Dual-stage feed drive concept

The dynamics of the drive stages, fine and coarse, will determine the way they work together complementary to achieve the maximum effect. Ball screw feed drives or LM feed drives may be used as coarse drive stages while PAs, magneto-strictive actuators or voice coils may be used as fine drive stages.

There are many considerations when it comes to designing a DSFD. These include:

- The travel range of the fine drive stage is relatively small and can be easily saturated.
- The DSFD system has an actuation redundancy. This means for a fixed target point there are infinite number of setpoint combinations. The system controller has to determine the best solution. The overall system should have consistent dynamics and this requires a research effort to determine how to split the setpoint between the two drive stages.
- The dynamic coupling between the two drive stages raises the option of treating the system as either two independent single input single output (SISO) systems, as in Ref. [20], or as a multi-input single output system.
- Geometrical constraints and tolerances of the mechanical components should be designed carefully to avoid losing the precision of the motion.

Regarding the consistent dynamics problem, two cases may be found depending on the bandwidth of the fine drive stage relative to the coarse drive stage. They are:

- The fine drive stage has much faster dynamics (i.e. much higher bandwidth) than the coarse drive stage.
- The fine and coarse drive stages have similar dynamics (i.e. no significant difference in bandwidth).

In the first case, the setpoint has to be split into two frequency components where the coarse drive stage tracks a nominal low frequency trajectory while the fine drive stage compensates for the tracking errors and responds to the high frequency component of the trajectory. This design criterion

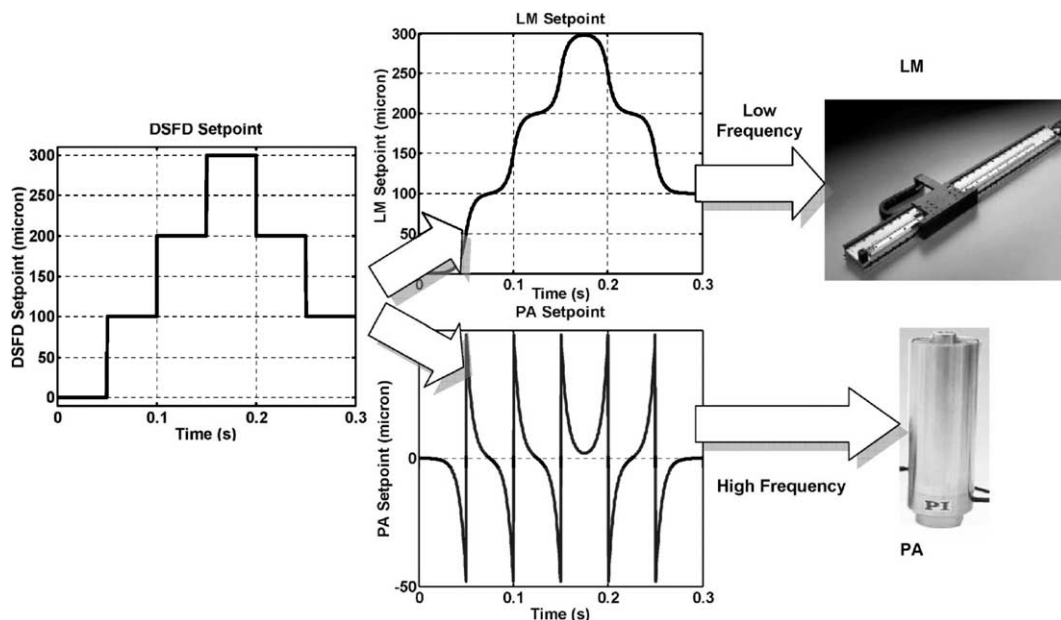


Fig. 1. Design criterion for high bandwidth DSFD.

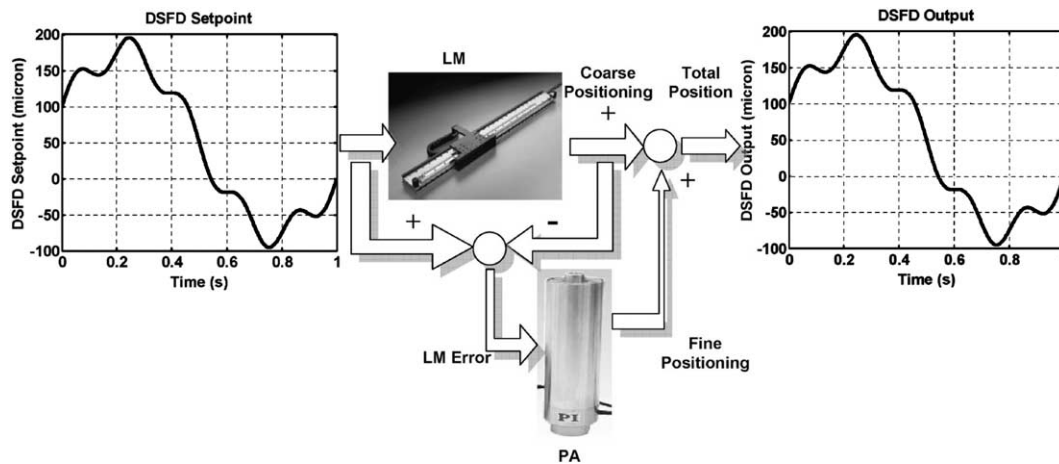


Fig. 2. Design criterion for high precision positioning DSFD.

is shown in Fig. 1 where the slow dynamics actuator is a LM, while the fast dynamics actuator is a PA. This case has applications when the overall bandwidth of the system is required to be increased as well as the precision. In the second case, as shown in Fig. 2, the error of the coarse drive stage is fed as the setpoint to the fine drive stage. This case has applications where the priority is the positioning precision.

4. Dual-stage feed drive design for a single-axis

4.1. Feed drive stages specifications

The DSFD system under study consists of two stages. The coarse feed drive stage is a LM, while the fine feed drive stage is a PA. The LM is a Baldor ironless core, three-phase, double sided, long secondary, and short primary type. It has approximately a 1 m travel, a 1 μm resolution glass scale encoder, and a sinusoidal three-phase amplifier. The motor continuous force is 90 N, peak force is 260 N and the mass of the moving carriage is 1.77 kg. The LM is controlled using a feedforward controller that is connected in series with a feedback controller in a two degrees-of-freedom (2DOF) structure to ensure excellent disturbance rejection and a fast response. A disturbance observer module is added to pre-compensate for model uncertainty and external low frequency disturbances. An anti-windup module is implemented to reduce the effects of amplifier saturation. This module adjusts the setpoint to keep the control command within saturation limits without causing a severe effect on the transient response. Feedforward compensators for force ripple and friction force are implemented to produce improved performance. The LM can achieve a bandwidth of 80 Hz experimentally.

The PA that is used in the system is made by Polytec PI and has push/pull forces of 12,500/2000 N and a 180 N/ μm static stiffness. The maximum travel distance of the actuator is $\pm 40 \mu\text{m}$. The actuator has a strain gauge (SG) position feedback sensor with 20 nm resolution. The PA is controlled using a multiple model-based switching controller.

This controller technique uses an iterative sequence to calculate the expected controller output (or amplifier input) based on the position error and the suitable model of the system. The control system consists of separate controllers that are designed for each linear model, and then a switching strategy selects the most suitable control command. Each controller is designed as a 2DOF controller plus an anti-windup module to deal with amplifier saturation. The PA bandwidth was determined experimentally to be 100 Hz. The detailed design and testing of the above mentioned controllers is presented in Ref. [21].

4.2. Single-axis PA flexure base design and analysis

When the positioning resolution is on the order of microns flexure bases are superior to conventional bearings [1]. A single-axis flexure base for the PA was designed to join the two feed drive stages. Fig. 3 shows the flexure base assembled with the PA. The flexure base is a frictionless and stick-free parallel-crank linkage that works based on the elastic deformation of a solid material. It eliminates all sliding and rolling actions. It also provides high stiffness and load capacity. The base material was selected to be Aluminum Alloy 1100-H14 to minimize the total moving mass carried by the LM for better acceleration performance. The material has the following mechanical properties: yield strength 95 MPa and Young's modulus of elasticity of 70 GPa. The dimensions of the elastic columns in the base were selected to permit the full range of expansion for the PA within the available pushing force of the actuator. For an 80 μm expansion a pushing force of 5000 N is required which is 41.6% of the PA capacity.

A finite element (FE) model was built for the flexure base. The model was used to perform the static and dynamic analysis for the base. Fig. 4 shows the stress distribution of the base due to a 5000 N static force. The maximum stress was calculated to be 65 MPa which represents a 1.5 factor of safety. A modal analysis was performed on the base to calculate the natural frequencies and mode

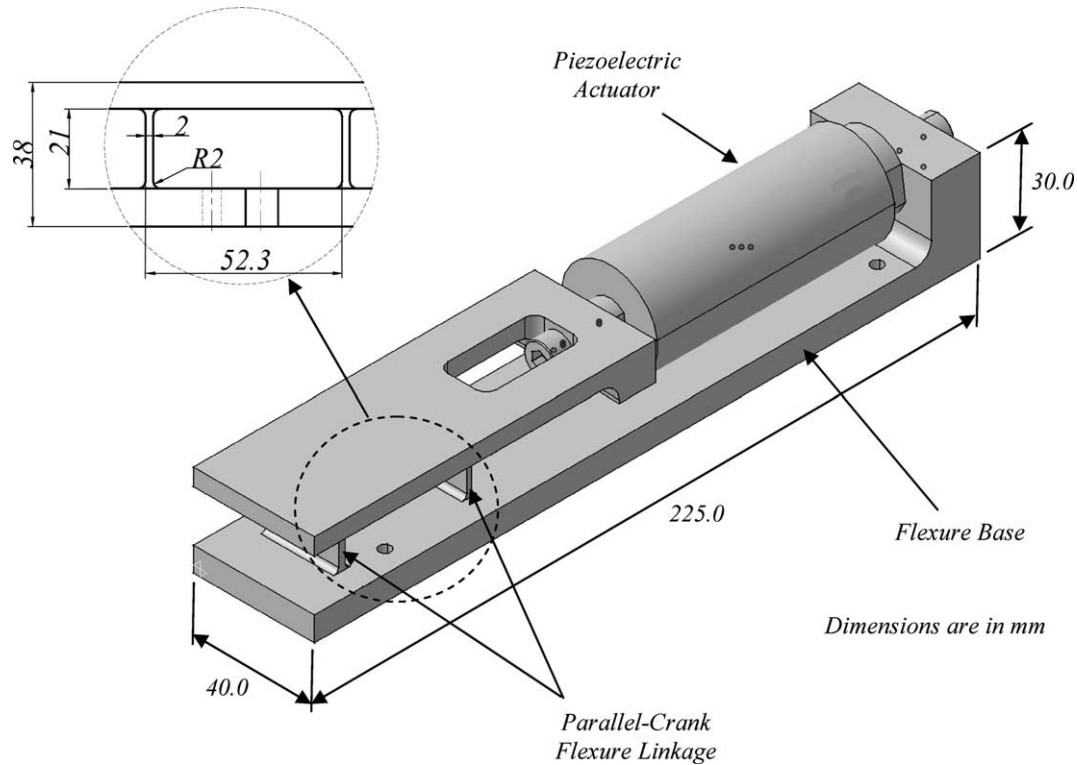


Fig. 3. PA single-axis flexure base.

shapes. The frequency range under study is 0–1000 Hz. This frequency range represents the expected maximum range of excitation force from a milling process. It is equivalent to a slotting operation with a spindle speed of 15,000 rpm and a four tooth end mill. No natural frequency occurred in this range; and the first natural frequency for the flexure base with the PA installed was 5000 Hz which is far beyond the excitation frequency.

The PA was assembled to the flexure base using two M8 bolts and the assembly is rigidly fixed to the LM moving

carriage using four M5 bolts and a 7.8 mm dowel pin. Figs. 5 and 6 show the single-axis DSFD assembly. The axis is theoretically capable of a 1 m stroke with 20 nm resolution.

4.3. Control of the DSFD

The control of the DSFD consists of two levels. The first is the feed drive stage level that deals with each drive stage separately. Within this level a SISO controller was designed

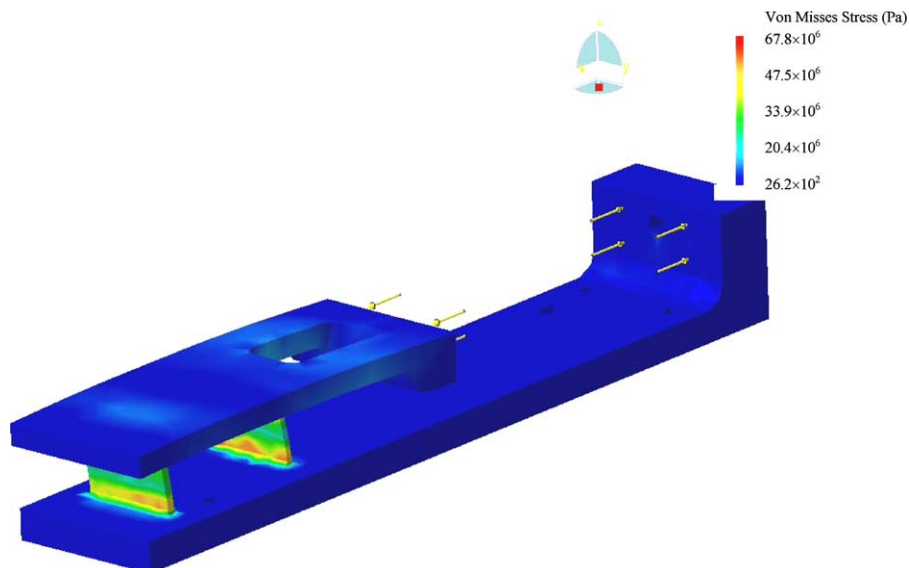


Fig. 4. Stress distribution at maximum static deflection.

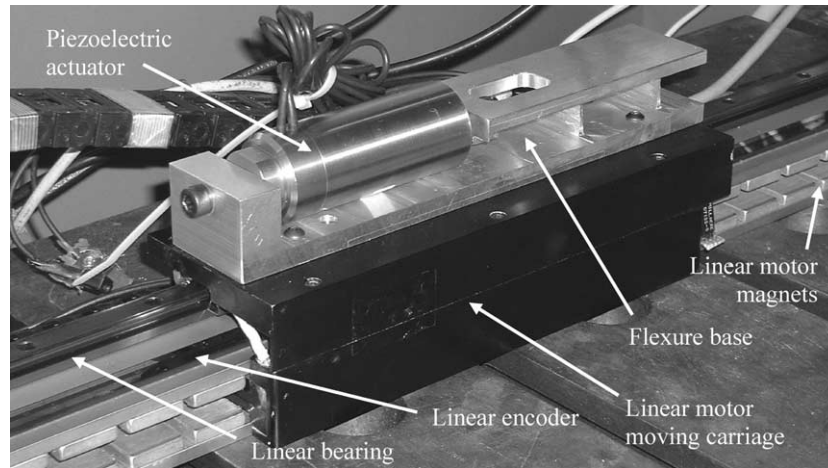


Fig. 5. Single-axis DSFD assembly.

for each drive stage based on stage dynamics. The controller input is the setpoint component for the drive stage while the output is a control command to the drive stage power amplifier. The second level is the setpoint handler. The setpoint handler takes the setpoint for the system and splits it into two components, one for each drive stage. The general idea of the first control level was discussed in Section 4.1.

The setpoint handler is by far the most important module in the DSFD system. It determines how the drive stages work together in a complementary way to improve the performance. As discussed before, there are two ways to split the setpoint between the drive stages: based on the frequency content and based on the positioning resolution. The relative speed of the dynamics of the drive stages is the determining factor. If there is much difference in bandwidth between the two drive stages then splitting the signal based on the frequency is the preferred option. When they have similar bandwidth, it is more effective to use the fine drive stage to compensate for the error of the coarse drive stage.

In our case, both the fine and coarse drive stages have very close bandwidths, 80 Hz for the LM and 100 Hz for the PA. The flow chart for the setpoint handler is given in Fig. 7. The coarse drive stage is commanded to follow the original

setpoint. When the error is less than a threshold value, the setpoint for the fine drive stage is made equal to the error. If the error is not less than the threshold value then the setpoint for the fine drive stage equals the product of the threshold and the sign of the error. In this way both drive stages are always working together to reduce the error. The only design variable is the error threshold. This should be made less than or equal to the maximum allowable displacement for the fine drive stage. It was determined experimentally that using the maximum displacement, i.e. $e_{\text{threshold}} = 40 \mu\text{m}$, was most effective with our PA. Experimental results with this setpoint handler are discussed in Sections 4.5 and 6.

4.4. Dynamic couplings between feed drive stages and with the machine base

The piezoelectric drive stage is dynamically coupled to the LM drive stage since the PA forces transmit to the LM. The PA forces consist of inertia forces from acceleration/deceleration of the PA moving mass and cutting forces. Dynamic coupling between the drive stages, and between the coarse drive and machine base, has to be studied because it may affect the tracking error and machined surface finish since the dynamics of

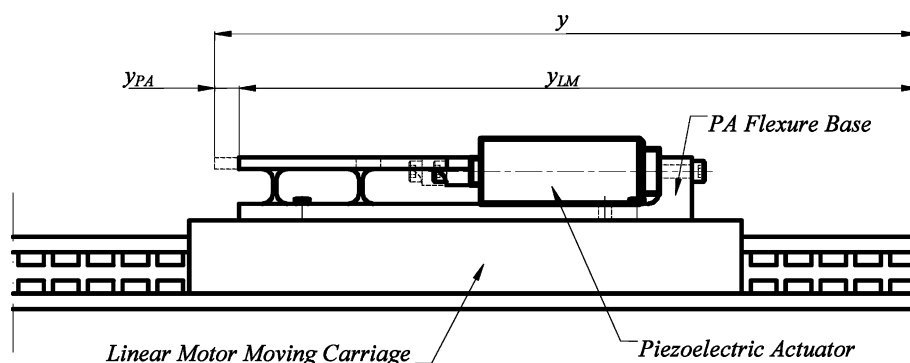


Fig. 6. Single-axis DSFD schematic.

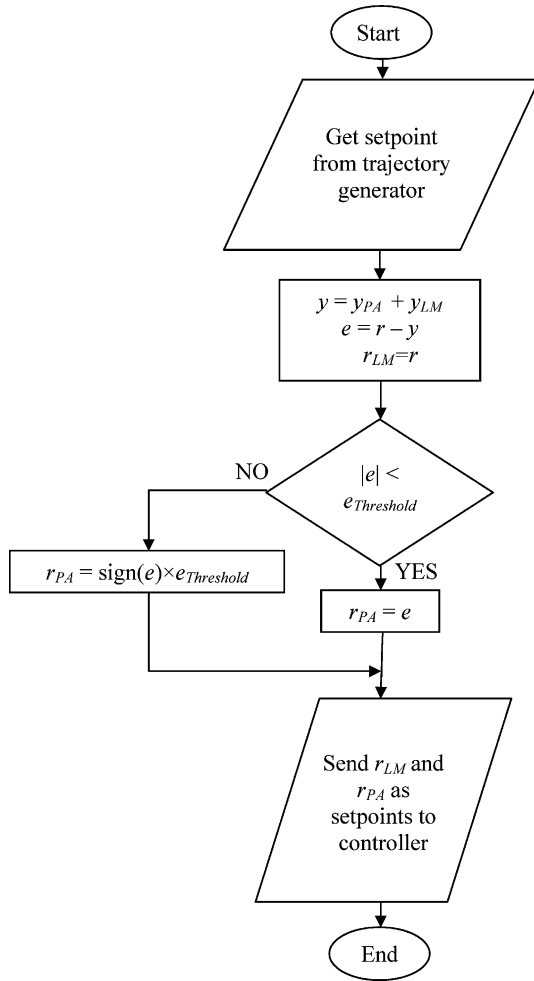


Fig. 7. Setpoint handler flow chart with overlap between the LM and the PA.

a coupled system are not the same as the dynamics of any two independent subsystems [22].

To determine the dynamic coupling, two transfer functions should be identified: the receptance of the LM, R_{LM} , and the transmissibility of the LM, T_{LM} . Receptance is the inverse of the dynamic stiffness of a feed drive and is the transfer function between the displacement of the drive and the disturbance force. The transmissibility for a feed drive is the transfer function between the transmitted force to the machine base and the external disturbance force (i.e. inertia force plus cutting force). Fig. 8 shows the DSFD model block diagram. It will be used to determine both the receptance and transmissibility functions for the LM. The following assumptions were made:

- The PA is relatively rigid compared to the LM.
- The setpoint to the LM is zero and the controller is responding just to the external disturbances.

4.4.1. LM receptance R_{LM}

LMs have no mechanical elasticities between the moving carriage and the stationary part. So the dynamic stiffness of a LM is completely dependent on the feedback controller settings. The feedback controller of the LM under study is a PID and can be written as in Eq. (1)

$$C_M(z) = \frac{5.35 \times 10^4 z^2 - 1.05 \times 10^5 z + 5.15 \times 10^4}{z^2 - z} \text{ V/m} \quad (1)$$

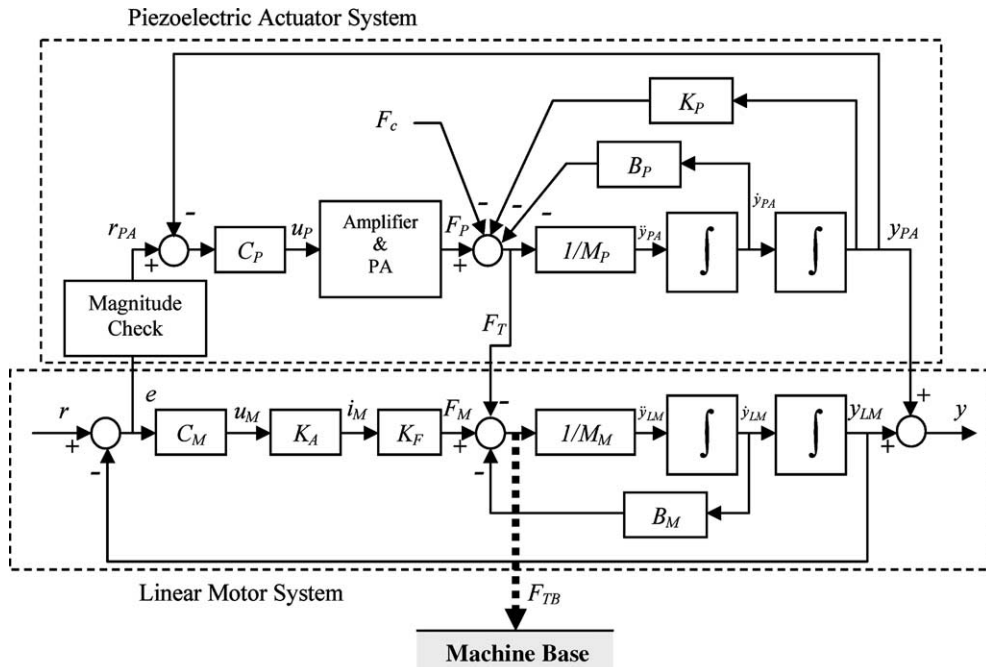


Fig. 8. DSFD model block diagram with feedback control modules.

The LM has the following parameters; $K_A = 0.4021$ A/V, $K_F = 45.6$ N/A, $M_M = 1.77$ kg, $B_M = 18.5$ ns/m and $T = 0.0005$ s.

By manipulating the block diagram in Fig. 8 and using backwards differencing to approximate the derivatives, the receptance transfer function $R_{LM}(z)$ which is $Y_{LM}(z)/F_T(z)$ can be calculated as in Eq. (2)

$$R_{LM}(z) = \frac{1.2348 \times 10^{-7} z^2}{z^2 - 1.87z + 0.8743} \text{ m/N} \quad (2)$$

Fig. 9 shows the magnitude of the frequency response of $R_{LM}(z)$. Some conclusions may be drawn from this plot. First, the curve shows no resonance peak. This means that the LM controller added sufficient damping. Second, the bandwidth of $R_{LM}(z)$ is 14 Hz with a static stiffness of 34.8 N/mm. This indicates that the LM is acting as a low pass filter and it will not be affected by high frequency disturbances. Lastly, the LM displacement due to maximum inertia disturbance force from the PA may be calculated as follows. The peak amplitude of the disturbance inertia forces expected at 14 Hz is about 0.3 N (for $M_P = 1$ kg and the maximum PA displacement amplitude of 40 μm). This would cause the LM to be displaced by approximately 9 μm . This amount of displacement is within the range of the PA for compensation. In case of higher displacement magnitude than the PA compensation range or when the coarse actuator contains resonance peaks, the control system must be redesigned to account for these effects. There are a few options. Firstly, F_T could be estimated and the estimate used for feed forward compensation. Secondly, a redundant PA could be added and used to damp the motion in the LM by counterbalancing the disturbance force.

4.4.2. LM transmissibility T_{LM}

The block diagram, Fig. 8, can be rearranged to extract the transmissibility function $T_{LM}(z)$ which is the ratio $F_{TB}(z)/F_T(z)$. This function gives a measure of how

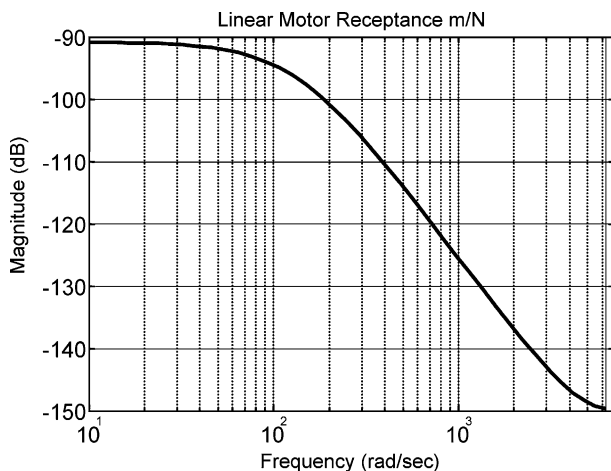


Fig. 9. Magnitude of the LM receptance function.

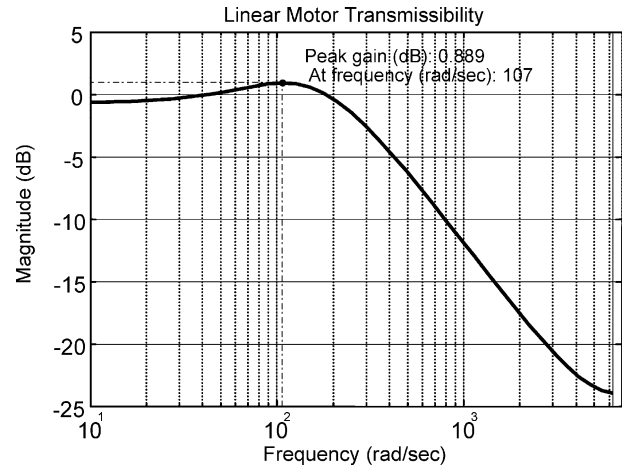


Fig. 10. Magnitude of the LM transmissibility function.

the fine drive is decoupled from the machine base dynamics. Eq. (3) represents the transmissibility function for the LM

$$T_{LM}(z) = \frac{0.12114z^2 - 0.1166z}{z^2 - 1.87z + 0.8749} \quad (3)$$

Studying the magnitude of the frequency response for $T_{LM}(z)$ shown in Fig. 10, the function has a peak magnitude of 1.107 at 17 Hz with a bandwidth of 50 Hz. At 50 Hz the maximum disturbance force amplitude due to the PA is about 3.5 N. So the maximum transmitted force to the machine base at 50 Hz is 3.4 N. This force value is very small and will not affect the machine base. In other applications where the fine drive transmits higher disturbance forces or magnitude peaks exist in the transmissibility function, a control technique called ‘reaction null space’ [23] can be applied to totally decouple the micro-actuator from the base dynamics and result in a zero reaction maneuver system [23].

4.5. Experimental results and analysis for the single-axis DSFD

In this section, the DSFD system is tested against a single-stage LM drive controlled by a model-based controller and by a PID controller to verify the effectiveness and the tracking characteristics. Sine wave and ramp inputs are used to compare the tracking error for the three systems, in Figs. 11 and 12, while step inputs are used to compare the steady-state error between the dual-stage and single-stage systems, in Fig. 13. The maximum tracking and steady state errors are summarized in Table 1. These results show that the DSFD is able to improve the tracking performance. In the case of sine wave the tracking error was reduced from ± 100 μm for a single-axis LM controlled by a PID feedback controller to ± 30 μm for a single-axis LM controlled using model-based control to ± 10 μm for the single-axis DSFD. The steady state error was improved by 60% using the single-axis DSFD over a single-axis LM controlled by

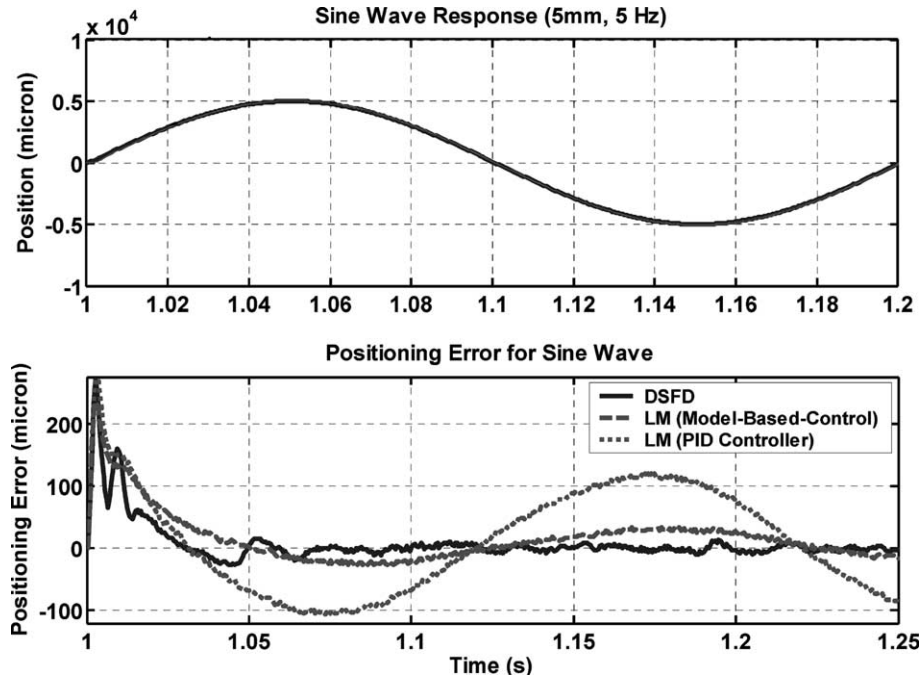


Fig. 11. Sine wave response and positioning error.

model-based control. It also achieved three times faster settling time for the step input (i.e. from 0.045 ms for the single-stage drive to 0.015 ms for the DSFD).

5. Design of two-axis dual-stage feed drive

This section presents for the design of a two-axis DSFD. The design employs two PAs and two LMs to produce the endpoint movements in two orthogonal directions.

5.1. Two-axis PA flexure base design and analysis

A two-axis flexure base is required for the implementation of two identical PAs as the fine actuators. It was designed based on the same concept as the single-axis flexure base and it has the same material properties. Fig. 14 shows an isometric view for the base assembled with the two PAs. The workpiece fixturing area (WFA) has dimensions of 150 mm $\times$ 150 mm.

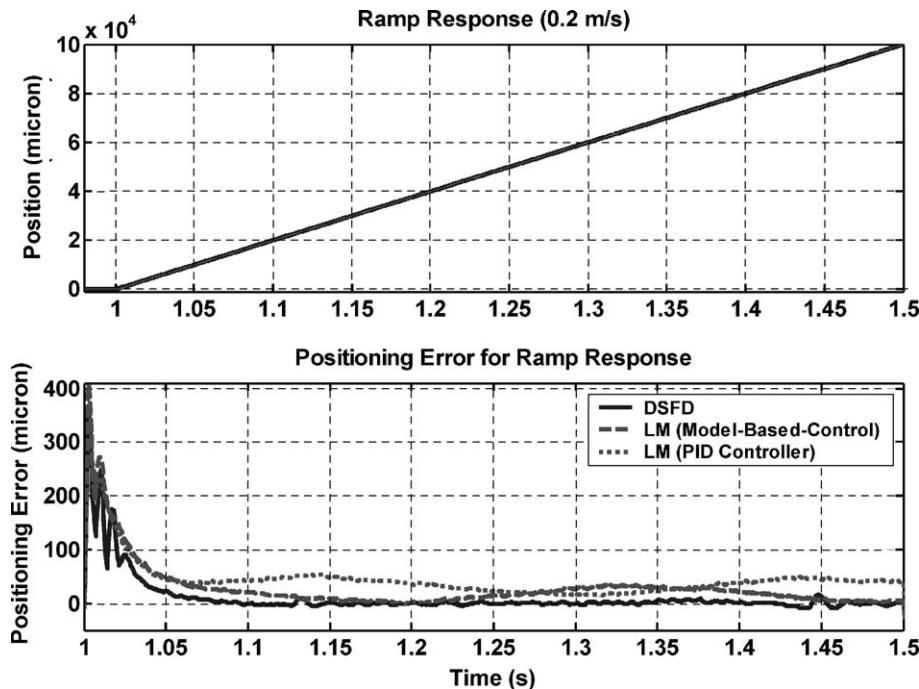
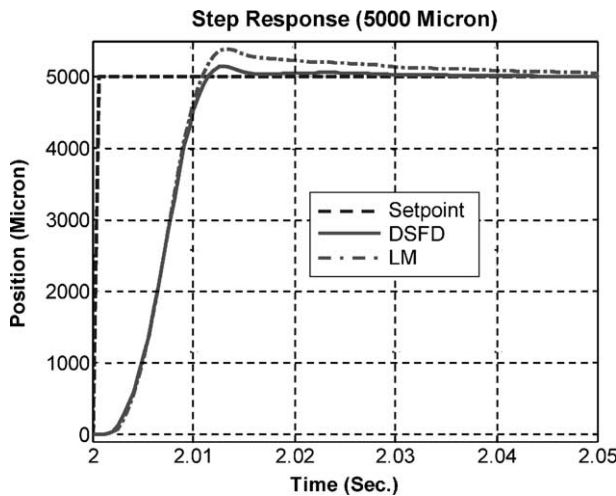


Fig. 12. Ramp response and positioning error.

Fig. 13. 5000 μ m step response.

The base was analyzed statically using FE analysis to check on the maximum stress value and the uniformity in displacements of the WFA in the X and Y directions. Fig. 15(a) demonstrates the stress distribution and deflections due to a fixed PA displacement of 80 μ m. The stresses reached a maximum of 26 MPa which is 27% of the material yield strength. No coupling between the two axes or rotating of the WFA was observed. The X and Y displacements were uniform over the WFA.

A dynamic FE analysis was required to get an estimate of the natural frequencies and the mode shapes of the flexure base within the range of the operating frequency. Fig. 15(b) and (c) shows the first two mode shapes in the range of 0–1000 Hz. The first natural frequency is about 650 Hz, while the second natural frequency is about 680 Hz. These two mode shapes are located away from the PA controller bandwidth but could be excited by a cutting process. The spindle speed and the number of teeth of the milling cutter must be selected to avoid these frequencies.

Table 1
Comparison in tracking performance between a single-stage LM drive and the single-axis DSFD

Test type	Feed drive	e_{tracking} (μ m)	e_{ss} (μ m)
Sine wave, 5000 μ m, 5.0 Hz, $v_{\text{max}} = 0.16$ m/s	LM (PID controller)	± 100	N/A
	LM (model-based control)	± 30	N/A
	Single-axis DSFD	± 10	N/A
Ramp, $v = 0.2$ m/s	LM (PID controller)	+80	N/A
	LM (model-based control)	+50	N/A
	Single-axis DSFD	± 10	N/A
Step, 5000 μ m	LM	N/A	–1 $\leftrightarrow$ +4
	Single-axis DSFD	N/A	± 1

5.2. Design features of the two-axis DSFD

Fig. 16 shows the design of the two-axis DSFD. Two LMs are connected orthogonally in series to form the two-axis coarse drive. The two-axis PA drive is mounted on top of the X axis LM. The design features a base plate to be mounted on an existing machine tool for testing purposes. The design also features high accuracy linear guides and linear bearings. The final resolution and the maximum speed are limited by the LM encoder specifications which are 1 μ m resolution and 1 m/s maximum speed in both X and Y directions.

6. Experimental testing in metal cutting

Since only one linear motor was available the two-axis DSFD design was not built. Instead a two axes PA flexure base was manufactured and assembled to a single-axis LM. The objective was to test the DSFD under real metal cutting conditions. The results of this testing will help to evaluate the performance of the DSFD compared to a single-stage feed drive. Fig. 17 shows the assembly of the two-axis PA flexure base to a LM. This DSFD was installed on a CNC three-axis milling machine. The DSFD was setup to feed the workpiece in the X direction, while the CNC machine fed the whole assembly in the Y direction. The Y axis PA was set to dwell.

Two groups of tests were performed, the objective of the first group was to mill a sharp corner and the objective of the second was to mill a sinusoidal profile. The cutting conditions were as follows: cutter diameter = 5 mm (four tooth high speed steel end mill), spindle speed = 12,800 rpm, feed in Y direction = 300 mm/min, depth of cut = 1 mm and duration of cutting = 30 s. For each group a comparison between the LM acting as a single-stage feed drive and the DSFD was established by monitoring the tracking error. Fig. 18 shows a sinusoidal profile cut using the DSFD.

Fig. 19 presents the experimental tracking performance of both the single-stage and dual-stage drives with sharp corner tracking. Only the data in the vicinity of the corner is plotted. The results demonstrate that the dual-stage drive is able to reduce the maximum tracking error by 38% and to reduce the average magnitude of the error by 39%. The oscillations in the tracking error were caused by the disturbance force due to the cutting process. The lower frequency component corresponds to the cutter runout frequency, while the higher frequency component corresponds to the tooth passing frequency. Fig. 20 shows the tracking performance with the sinusoidal profile and demonstrates an 83% reduction in the maximum tracking error and 63% reduction in the average magnitude of the error using the DSFD.

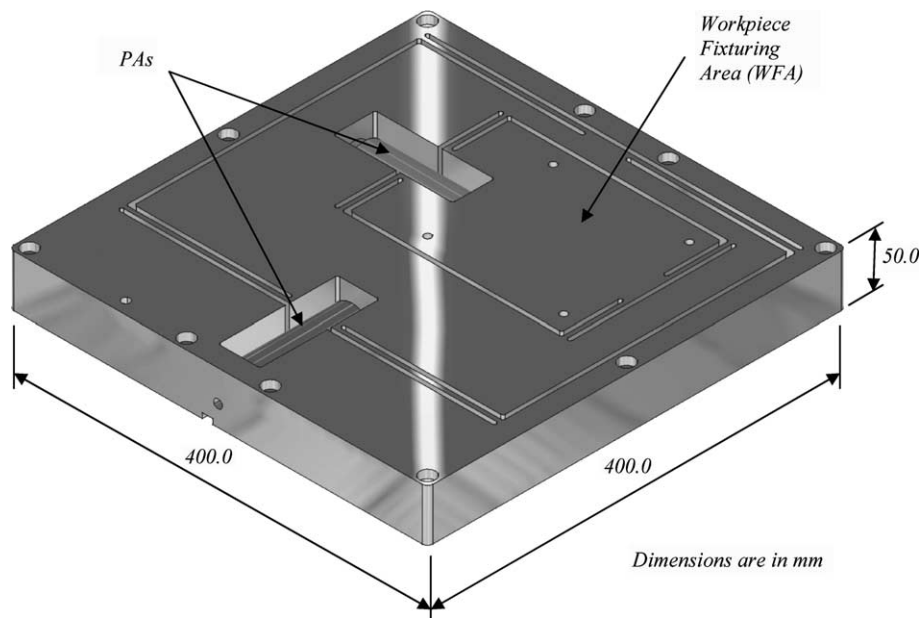
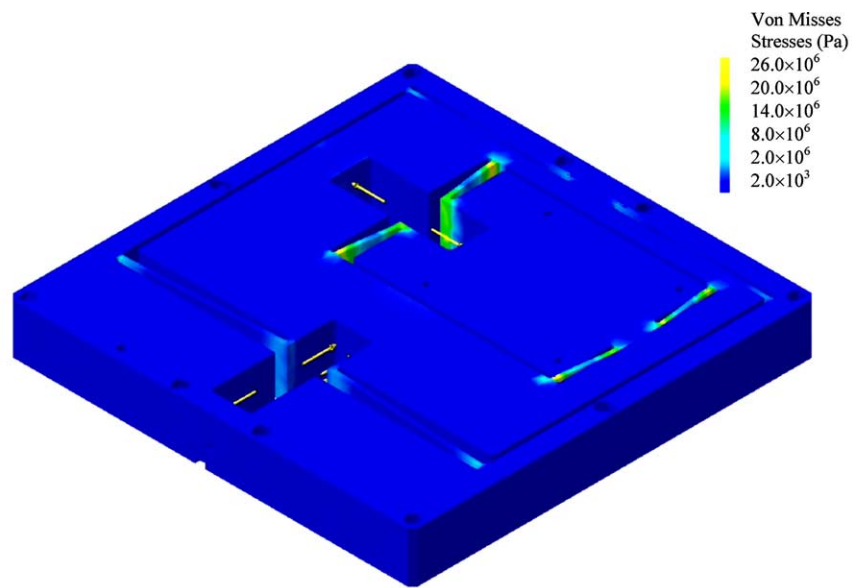
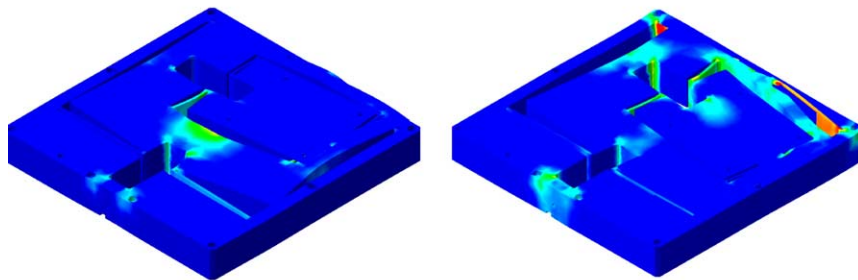


Fig. 14. Two-axis PA flexure base assembly.



(a) PA flexure base static analysis.



(b) First mode shape at 650 Hz.

(c) Second mode shape at 680 Hz.

Fig. 15. Two-axis PA flexure base static and dynamic analysis.

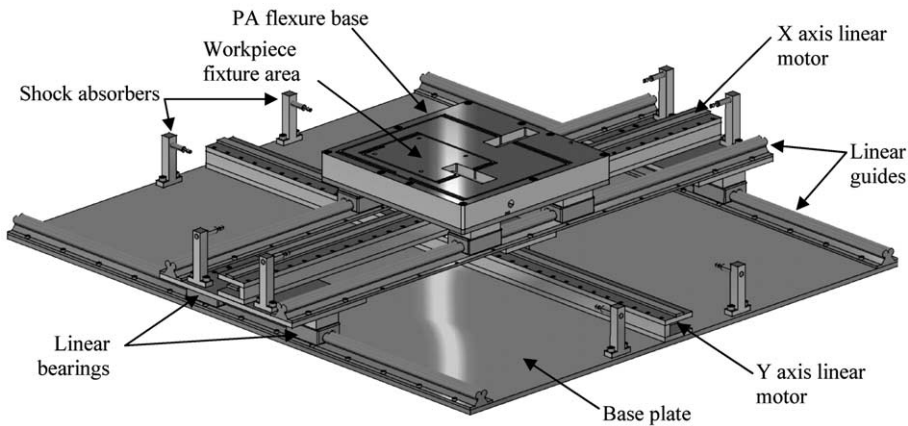


Fig. 16. DSFD design in two dimensions.

7. Conclusions

This paper described the design and control of DSFDs for precision machining. The issues involved in DSFD design were analyzed and solution approaches presented. The control algorithm design considers the relative bandwidths

of the two drive stages and ensures that they work together in a complementary way. The dynamic couplings between the two stages, and between the coarse stage and the machine base, were analyzed and were determined to be unlikely to affect the tracking error or machined surface finish. The designs for single-axis and two-axis DSFDs were presented. Both designs utilize flexures to provide precise frictionless motion. FE analysis was used to predict

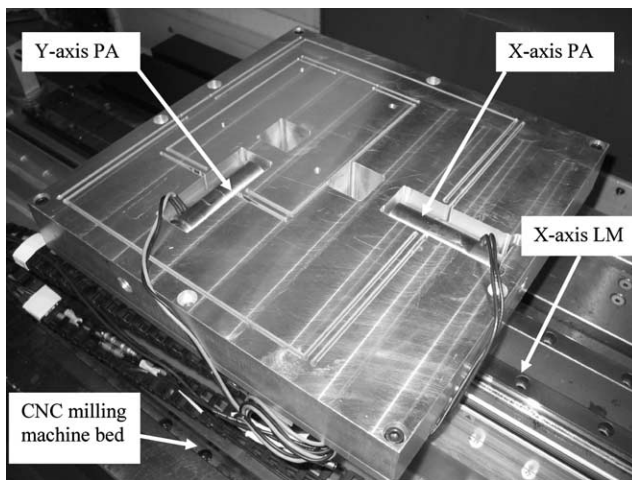


Fig. 17. The DSFD installed on three-axis milling CNC machine.

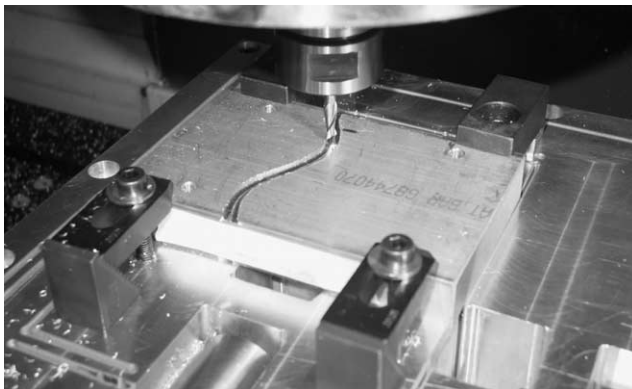


Fig. 18. Sinusoidal profile cut using the DSFD from Fig. 17.

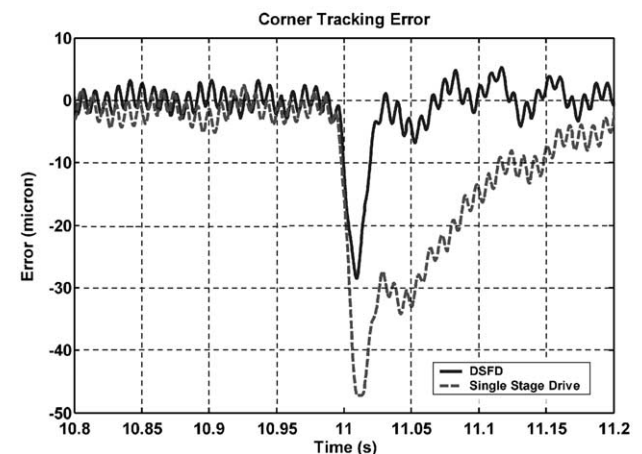
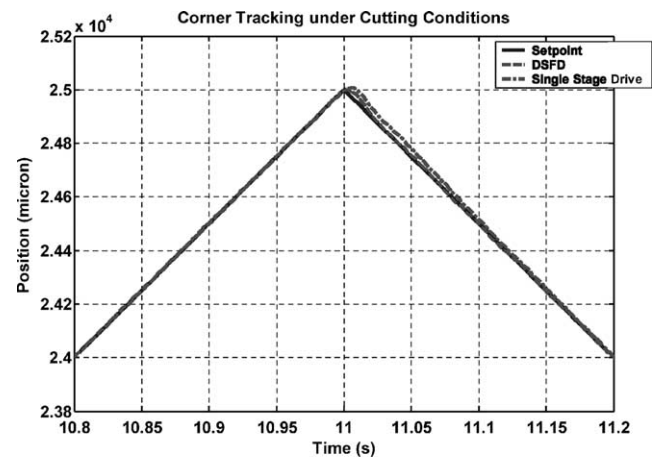


Fig. 19. Experimental results from corner tracking while cutting.

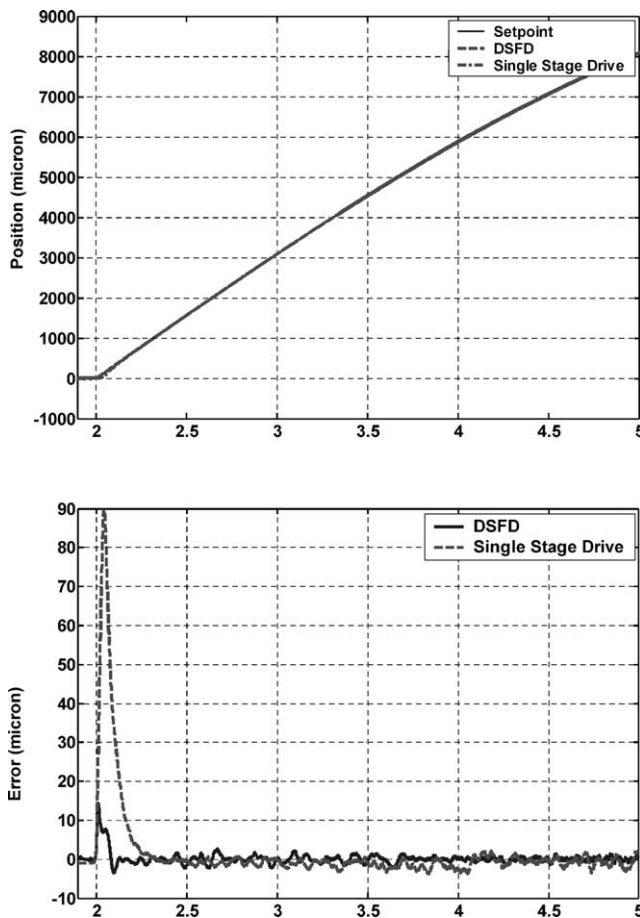


Fig. 20. Sinusoidal profile tracking while cutting and the resulting tracking error.

the static and dynamic flexure performance and their relationship to the milling process requirements. In experiments with the single-axis DSFD, the tracking error was reduced by about 75% in comparison to a similarly controlled LM drive. Milling tests were conducted with the second DSFD. In sinusoidal profile cutting the maximum tracking error was reduced by 83% and the average magnitude of the error was reduced by 63%. In sharp corner cutting the maximum tracking error was reduced by 38% and the average magnitude of the error was reduced by 39%.

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